

README File and Help File to Run the Program

Project Title:

Synthetic Facial Image Database Generation and Validation for Facial Palsy

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Project Description:

This project develops a web-based system for organising, analysing, and validating synthetic facial images generated for facial palsy research. The system allows images to be loaded under broad intended generation classes such as Healthy, Mild_Palsy, Moderate_Palsy, and Severe_Palsy. Each image is then analysed using facial landmark detection and asymmetry metrics to estimate the visible severity of facial palsy.

The final output classifies images into six severity grades inspired by the House-Brackmann scale:

- * Grade I - Normal
- * Grade II - Mild
- * Grade III - Moderate
- * Grade IV - Moderate-Severe
- * Grade V - Severe
- * Grade VI - Total Paralysis

Synthetic Image Generation to Run the Model on Kaggle

I have included the notebook *lora_training_and_imageGeneration.ipynb*, which contains the full implementation for the LoRA training process and the synthetic facial palsy image generation pipeline. The notebook documents how the base diffusion model was fine-tuned using LoRA and how the trained model was later used to generate synthetic facial images across the selected facial palsy severity classes.

Project Structure:

- * app.py: Main Flask application file
- * config.py: Configuration settings
- * src/metrics.py: Facial landmark extraction and asymmetry analysis functions
- * static/css/: Stylesheets for the user interface
- * static/js/: JavaScript files for frontend actions
- * templates/: HTML pages for the web interface
- * sample_images/: Small sample images for testing the system
- * requirements.txt: Python dependencies needed to run the project

How to Run the Project:

1. Create and activate a Python virtual environment.

2. Install the required dependencies:

```
pip install -r requirements.txt
```

3. Run the Flask application:

```
python app.py
```

4. Open the local web address shown in the terminal, usually:

```
http://127.0.0.1:5000
```

5. Load or select images and use the analysis/validation function to view predicted severity grades and asymmetry scores.

Important Notes:

- * The selected class represents the intended generation class, not the final ground-truth label.
- * The final predicted grade is produced by the landmark-based validation system.
- * Some images may be flagged as “Needs Review” if the predicted grade does not match the intended class or if possible image artefacts are detected.
- * The full generated dataset and model weights are not included in this submission due to file size limitations.
- * A small sample image folder is included only for testing and demonstration.